Final Project Proposal

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Introduction

THE UNIVERSITY OF TEXAS AT DALLAS
Office of Facilities & Economic Development

- F&ED's Mission: To assist UTD in becoming an 'economic convener' for the metroplex's growth by developing opportunities between the university and various public and private partners.
- Monthly: Produce reports of DFW's labor market. Composed mostly of data hosted by the Bureau of Labor Statistics (BLS).

U.S. BUREAU OF LABOR STATISTICS

Problem –> Project Purpose

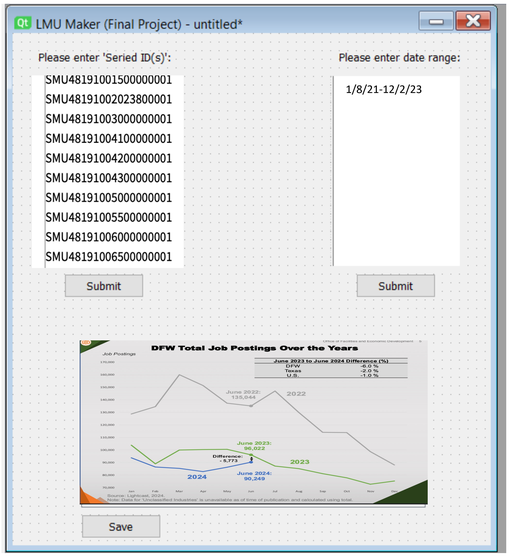
Copy the table from G2:M15 below.

Industry	May. 2023 Employment	Mar. 2024 Employment	Apr. 2024 Employment	May. 2024 Employment	Change, May. 2023 to May. 2024 (%)	Change, Apr. 2024 to May. 2024 (%)	
Mining, quarrying, and	104,200	109,000	109,000	110,200	5.76%	1.1%	Mining, quar
Construction	140,000	141,400	142,100	142,800	2.00%	0.5%	Construction
Manufacturing	305,200	309,000	309,900	311,100	1.93%	0.4%	Manufacturin
Wholesale and retail t	628,800	628,500	630,800	633,800	0.80%	0.5%	Wholesale a
Transportation and uti	261,500	263,900	263,300	265,600	1.57%	0.9%	Transportati
Information	92,800	91,000	90,700	90,500	-2.48%	-0.2%	Information
Financial activities	365,100	367,400	369,600	371,700	1.81%	0.6%	Financial ac
Professional and busi	778,100	767,500	773,600	772,300	-0.75%	-0.2%	Professional
Education and health	504,000	513,700	518,200	516,200	2.42%	-0.4%	Education a
Leisure and hospitalit	426,800	423,400	430,800	435,600	2.06%	1.1%	Leisure and
Other services	135,800	142,200	143,500	143,400	5.60%	-0.1%	Other servic
Government	476,000	488,300	490,800	493,100	3.59%	0.5%	Government
Total nonfarm	4,218,300	4,245,300	4,272,300	4,286,300	1.61%	0.3%	

- Use of outmoded pipeline in office.
- Public data-sets manipulated, and visuals designed, manually via MS Excel.
- Human-prone wrangling errors necessitates thorough checking processes.
- Other deadlines impacted.

GUI Interface Template

Use flow will involve a user entering a set of series IDs and a date range. A graph will be given as output. An example of what the final application might look like:

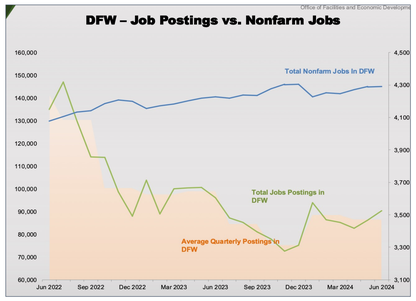


Inputs and Outputs

- Data series designators (SIDs) of facets of the regional economy will be entered by analyst within a text-field on left-side of app window. E.g., employment in the manufacturing sector would be entered as 'SMU48191001500000001'.
- Date range(s) will be entered within another text field next to the 'Series ID' field.

Enter series id(s) below:

SMU48191001500000001
SMU48191002023800001
SMU48191003000000001
SMU48191004100000001
SMU48191004200000001
SMU48191004300000001
SMU48191005000000001
SMU48191005500000001
SMU48191006000000001
SMU48191006500000001



- Targeted Metrics: Total employment and job posting changes in the metroplex. Filtering data to display common trends (a.k.a. level-changes over time) will be accomplished by 'Pandas'.
- Graphs of trends over time-span will be displayed in display window.
- Options to download produced images in '.png' format will be accessible via button below display segment.

End Product and Solution Methodology

- Design a standalone application in Python with a GUI interface that will allow automation of data processing and visuals production for producing LMU (Labor Market Update) graphs.
- Data will be retrieved via Python instantiated code pulling from BLS API. Data will be filtered with 'Pandas' and plots made with 'Matplotlib'. GUI elements will be designed with 'Qt Designer'.

Resulting Benchmarks

- Implement pythonic practices in development of novel application.
- Improve work-flow efficiency by about 25% per month.
- Introduce automation practices to analysis pipeline.
- Help make the case for the **DFW** area more efficiently.